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Residual stress formation and control in wire arc additive manufacturing: Mechanisms, characterization, and mitigation strategies

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Abstract Wire arc additive manufacturing (WAAM) has emerged as a competitive directed energy deposition (DED) technique for producing large-scale metallic components with high deposition rates and low material wastage. This direct energy deposition AM processes can either be gas metal arc welding or plasma arc welding. Given the freedom of attaining complexities of the geometries, being one of the few metal additive manufacturing processes and having a high deposition rate, it can be used in various applications that require intricate part preparation and critical areas like satellite systems, space explorations and high-performance vehicles. The materials primarily include alloys of aluminum, steel, copper or nickel.

Reported results consistently show that higher heat input and slower travel speeds intensify tensile longitudinal stresses, whereas optimized inter-pass temperature control and oscillation strategies help reduce gradients and promote microstructural stability. Advanced characterization techniques—including neutron diffraction, X-ray diffraction, ultrasonic methods, and magnetic Barkhausen noise—are compared in terms of penetration depth, accuracy, and suitability for WAAM components. Mitigation strategies such as heat treatment, rolling, laser shock peening, and controlled cooling are evaluated based on their demonstrated ability to relax residual stresses and refine microstructure. The variable heating and cooling of the sample causes development of the residual stresses. The residual stresses cause the material to fracture and fail which leads to the failure of the machine. The samples can be tested for various mechanical and thermal properties.

Simulated results can further be characterized by non-destructive methods for understanding the microstructural effects and X-ray diffraction for studying the grain structure of the samples prepared. This work consolidates current understanding of residual stress behavior in WAAM and outlines critical insights needed to guide process optimization, improve component reliability, and support the broader industrial adoption of WAAM for aerospace, automotive, and high-performance structural applications. Based on this study, the optimum sample selection methodology can be decided and further examination can be done for suitable applications.

1. Introduction

Additive manufacturing has transformed modern manufacturing by enabling the production of geometrically complex structures that were previously unattainable. Adin et al. recent advancements in additive manufacturing (AM) from 2021 onward have significantly expanded its industrial applicability by enabling complex geometries, material efficiency, and near-net-shape fabrication. Among various AM techniques, directed energy deposition (DED) processes such as wire arc additive manufacturing (WAAM) have gained substantial attention due to their high deposition rate, low material wastage, and suitability for large-scale metallic components. It provides us with a wide range of materials which can be combined and also various forms in which the material can be used and different techniques that are used to put the 3D model into shape. The

composite metals can be combined like Fe-Ni, Fe-Cu, and Inconel-825 with Co and many more. Recent investigations from 2023–2025 further demonstrate the advancements in thermal modelling, monitoring, and microstructural control in WAAM processes [1-3]. These developments have improved understanding of heat accumulation, bead morphology, and residual stress evolution. These materials may be employed in powder or wire form, depending on the process, with laser-based powder bed fusion and wire-fed WAAM being the most common approaches. WAAM is utilized in producing large metal parts using its high deposition rates of the material and ability to combine various materials. WAAM allows production of a large-scale metal parts in a considerably less amount of time due to its high deposition rate of metal wires as feedstock. Wire arc additive manufacturing uses an electric arc to heat and melt the feed stock material and deposition according to the design of the component [4]. As shown in Fig. 1, the torch carries the wire of the feedstock and the arc is produced using the current. Recent reviews on AM processes emphasize that mechanical properties of additively manufactured components are strongly influenced by process parameters, thermal history, and layer-wise solidification behavior, which often result in anisotropic microstructures and residual stresses. Post-processing strategies such as heat treatment, surface finishing, and stress-relief techniques have therefore become critical to improving structural integrity, fatigue performance, and dimensional accuracy of AM components. This melts the wire and it is deposited layer over layer to make the component [5].

WAAM is considered more cost effective than other powder-based processes as it uses a wire-based feedstock material and the metal wires are more easily available and safe to handle. This also reduces the overall cost of the part production even though the surface finish is not good, which adds the machining process after the sample is made. Many applications of this technology use a buy to fly ratio which is nothing but a ratio of raw material to final product. A low buy to fly ratio is preferred in the aerospace, automobile and other critical industries as the material wastage is very less [6]. A major challenge that occurs during the manufacturing of the part is residual stresses. Since the deposition rate is high and the heat input to melt the metal is also very high, the distortion generated in the parts is more. The optimized part can be produced by varying the different processes parameters like travel speed, wire feed rate, working distance, arc voltage, and deposition strategy or path planning [7]. In addition, improved sensing and in-process control strategies—such as embedded thermal sensors, digital-twin frameworks, and machine-vision monitoring—are now emerging as essential tools for high-quality WAAM fabrication [8-10]. These advancements reinforce the importance of analyzing residual stress behavior, as addressed in this review. Based on the various arc-based technologies used to melt the metal, WAAM is categorized into three categories as shown in Fig. 2.

Gas tungsten arc welding (GTAW)–based WAAM:

In gas tungsten arc welding, the arc is produced between the tungsten and the substrate. This arc melts the feeder wire which is fed from side. It uses a *non-consumable tungsten electrode* to create a stable arc while the filler wire is fed separately. It offers excellent control over heat input, produces clean deposits, and is suitable for high-quality but lower-deposition-rate WAAM builds.

Gas metal arc welding (GMAW)–based WAAM:

A consumable electrode method which is gas metal arc welding is the one where generated arc melts the wire and is then deposited on the substrate employs a *consumable wire electrode*, which acts as both heat source and filler material. GMAW provides the highest deposition rates, good automation compatibility, and is the most widely adopted WAAM process for large components. In addition, recent work on post-processing of fused deposition modelling (FDM) and metallic AM parts demonstrates that treatments such as annealing, surface modification, and peening significantly enhance microstructure uniformity, mechanical strength, and surface quality. These studies collectively indicate that integrating optimized process parameters with suitable post-processing routes is essential for achieving reliable performance in industrial AM applications. However, despite rapid progress, challenges such as residual stress formation, microstructural heterogeneity, and limited in-situ characterization methods remain critical barriers to widespread adoption, particularly in multi-material WAAM systems.

Plasma arc welding (PAW)–based WAAM:

In plasma arc welding, the arc is restricted between the water-cooled nozzle and non-consumable tungsten electrode. The inert gas flowing in the torch gets ionized and converted into

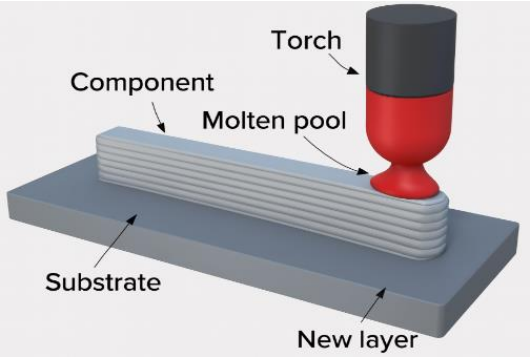


Fig. 1. A pictorial representation of wire arc AM process.

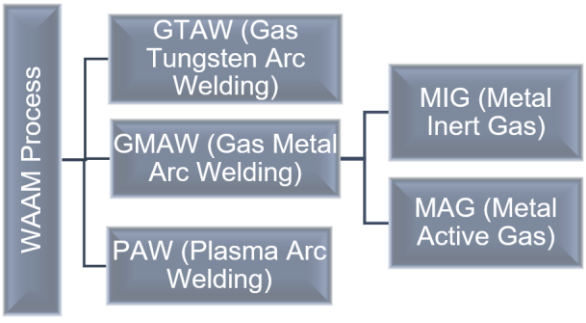


Fig. 2. Classification of WAAM.

plasma. It uses a constricted, high-energy *plasma jet* to melt the material, providing deeper penetration and a highly focused heat source. PAW offers improved stability and precision over GTAW, making it suitable for controlled, high-energy WAAM applications.

2. Research methodology

To provide a comprehensive and an unbiased assessment of the residual stresses generated in the samples made using wire arc additive manufacturing process, the methodology used consisted literature collected from major scientific studies indexed in manufacturing, additive manufacturing publications like Scopus, SpringerLink and Elsevier. The studies involving WAAM and the articles that studied various processes parameters were selected for inclusion. Later, the studies focusing on residual stress generation and post processing techniques were selected. This helped in establishing a coherence between the literature and analysis carried out by the authors in different studies. The flow chart below shows the flow of this review in order have an understanding of this study. The Fe–Ni alloy system was deliberately selected due to its pronounced ferromagnetic behavior, which is necessary for reliable Barkhausen noise–based residual stress evaluation. Additionally, Fe–Ni wires were chosen based on material availability and proven suitability in WAAM studies, where similar alloys exhibit stable melting behavior and predictable thermal response. The process parameters discussed in this study represent commonly adopted WAAM settings intended to generate controlled variations in bead geometry and thermal cycles, enabling systematic evaluation of residual stress effects.

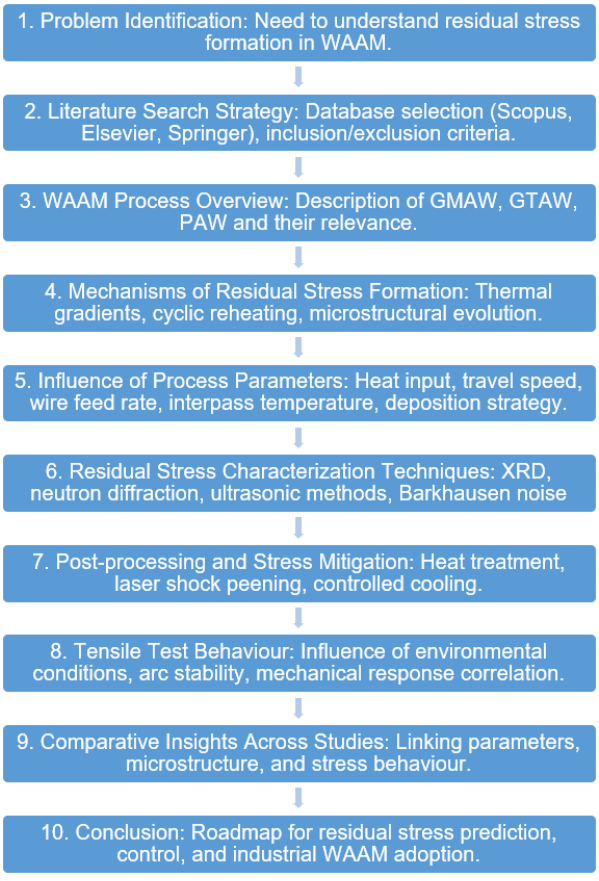


Fig. 3. Flow chart showing the methodology of this review.

3. Characterization methods for residual stresses in WAAM

Residual stresses generate due to lot of energy exchange caused by rapid heating, material deposition, cooling and reheating of the previous layer and cooling again. Residual stresses are an inherent characteristic of wire arc additive manufacturing (WAAM) due to the highly localized and repetitive thermal cycles involved in the layer-by-layer deposition process. These stresses originate primarily from non-uniform heating and cooling during arc melting and solidification. When the electric arc melts the feed wire and substrate, a steep temperature gradient is generated between the molten pool and the surrounding solid material. As the molten metal cools and contracts, it is constrained by the underlying solid layers, leading to the development of tensile residual stresses within the deposited region and compressive stresses in adjacent areas. The residual stresses vary based on the variation of different process parameters like current, material composition, scanning speed, probe height, and heat. Depending upon the length, the residual stresses can be classified as type I, type II, and type III. The type I residual stresses are caused due to the changes in the boundary conditions or process parameters. These stresses are the reason for the deformation of the parts. The type II residual stress are caused due to the unorganized grain structure of the specimen. The type III residual stress occurs due to the atomic level impurities, porosity and dislocations at nano level. The pie chart in Fig. 4 illustrates the relative importance of different factors causing residual stress in WAAM builds, based on the qualitative impact [11-13].

It is observed that the residual stresses caused are due to the temperature gradients and the properties like anisotropic grain structure from directional solidification of the materials being

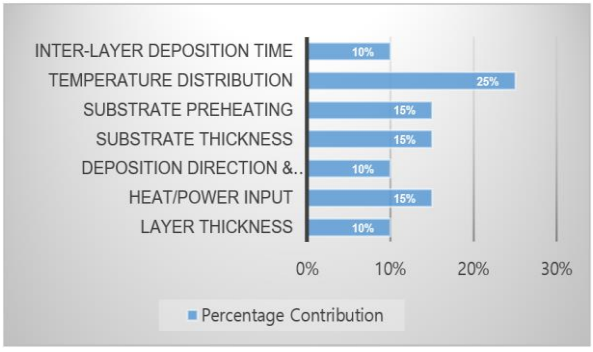


Fig. 4. Factors causing residual stress in WAAM builds.

Table 1. Process parameters.

Parameter	Details
Welding method	Gas tungsten arc welding (GTAW) using KEMPPI® MaterTig MLS 3000
Deposition current	140 A
Arc length	3.5 mm (tungsten tip to workpiece)
Deposition voltage	~13.1 V (during actual deposition)
Travel speed	95 mm/min
Filler wires	GoodFellow® 99.9 % purity Fe (FE005180) and Ni (NI005171), 1 mm diameter
Deposition alloy	Fe–Fe_3Ni functionally graded material (FGM)
Thermal control setup	Thermal insulated box with ceramic heating blanket
Interpass temperature	200 °C
Temperature monitoring	K-type thermocouple, 1 mm from build-up surface
Purpose of temp control	Prevent collapse from thermal accumulation and avoid cold cracking

used. One of the primary mechanisms responsible for residual stress formation in WAAM is thermal shrinkage mismatch. During deposition, the top layer experiences rapid cooling and contraction while the previously deposited layers remain relatively rigid, preventing free deformation. This constraint induces longitudinal tensile stresses along the build direction and transverse stresses across the bead width. Repeated deposition cycles further intensify this effect, resulting in stress accumulation throughout the build height. When the material is deposited layer by layer, the restricted thermal expansion and contraction results in residual stresses. These stresses vary at different positions in the deposited material. The tensile stresses are observed mainly at top and bottom in the longitudinal direction whereas the compressive stresses are observed in the transverse direction. The geometry of the samples also affects the stresses generated in it. Curved samples show tensile hoop stresses whereas flat ones show compressive stresses. Axial stress tends to be tensile at top/bottom and compressive internally [14]. During the deposition of material on the substrate, it is champhered and then the material is deposited. During this process, a lot of changes occur thermally, mechanically and metallurgical causing the generation of residual stresses. Another significant source of residual stress is the cyclic thermal loading associated with multi-layer deposition. Each newly deposited layer reheats the underlying material, causing repeated expansion and contraction cycles. These thermal fluctuations lead to plastic deformation and strain hardening, which contribute to the development of residual stress gradients within the component. The magnitude of these stresses is strongly influenced by process parameters such as heat input, interpass temperature, deposition speed, and layer thickness. To avoid the failure of parts, assemblies are clamped to avoid the strain but the residual stresses are still more. Authors observed that the reduced energy input per unit length will reduce the temperature gradient and hence the residual stresses. Authors also observed that the heat treatments also help in reduction of the stresses. It was concluded that the pre and post processing is required to reduce the residual stresses [15-19]. Authors used a bi wired GTAW setup to study the residual stresses in Fe-Ni alloy. Recent studies from 2023–2025 have contributed significantly to process parameter optimization in WAAM. The schematic representation is shown in Fig. 5.

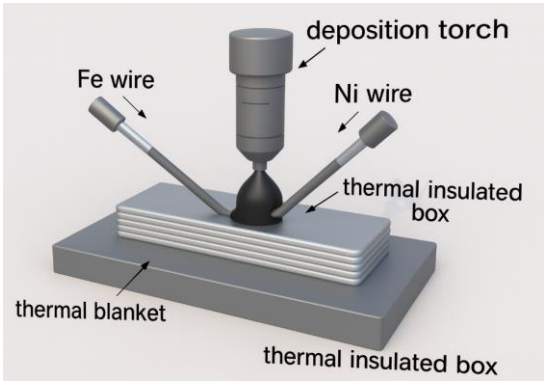


Fig. 5. Twin wire GTAW setup.

The details of the mechanism are as given in the Table 1.

Optimization of travel speed and wire feed rate for nickel alloys has been shown to directly influence internal stress profiles and porosity distribution [20, 21]. After the sample is observed using SEM, microstructural differences are observed at the top and bottom sections. The top shows columnar structure whereas the bottom layers show the cellular structure. It was observed that the heat treatment produces an alloy which is ductile and smooth. The residual stresses generate during the cooling of the material. The convective flow of the melt pool contributes less to the stress generation but it does affect the microstructure of the sample. Hence, during the stress calculation, the convective flow of the heat in the melt pool can be neglect in order to obtain smooth calculations of the heat equations. Authors observed that the residual stresses are generated when the cooling starts. When the clamps are released, the material deforms and stresses are released in the sample. The amount of stress generated for different weld lengths and pattern showed that the stresses generated can be reduced if the weld lengths are shorter [22]. Authors observed that the generation of residual stresses are less in the build direction which means

that the travel speed of the torch has less effect on the stress generation. However, the time taken to build the subsequent layer has a major effect on stress generation along the transverse direction and along the build direction. If the dwell time is reduced, the stresses are more evenly spread out and the stresses are found in the deeper side of the sample [23]. During the manufacturing of parts using WAAM, the process parameters play very vital role along with the geometry of the part, while considering the residual stresses. It is summarized in Table 2. Mechanical constraint conditions also contribute significantly to residual stress formation. The rigid fixation of the substrate plate restricts thermal distortion during deposition, leading to high tensile stresses near the substrate–deposit interface. As the build height increases, stress redistribution occurs, often resulting in distortion, warping, or delamination.

Overall, residual stresses in WAAM arise from the combined effects of thermal gradients, repeated heating cycles, metallurgical transformations, and mechanical constraints. Understanding these mechanisms is essential for optimizing process parameters, selecting suitable post-processing treatments, and ensuring the structural integrity and dimensional stability of WAAM-fabricated components.

Travel speed while depositing the material is one of the significant parameters as it affects the material being deposited. The greater travel speed will affect the surface roughness, insufficient material deposition and smaller bead height. This will result in the defects on the surface as well as the improper phase transformations. A sample with a) TS: 15 mm/s & power: 2500 W, b) TS: 5 mm/s & power: 4500 W, c) TS: 5 mm/s & power: 2500 W was prepared and the one with slow deposition speed showed a smoother surface [24-29].

Wire feed speed also affects the geometry and the surface properties of the bead height and width. An increased feed will increase the height and width but the surface roughness will also increase. If the feed rate is more than the melting rate, it will lead to wire expansion, arc instability, and poor droplet transfer which will lead to low stacking or rough surfaces. The torch angle also affects the geometry of the bead as being kept at a small angle other than perpendicular, leads to stronger bead geometry by avoiding the molten metal sagging [30-33]. Increasing weld current widens the bead but can reduce penetration depth due to higher forces (centrifugal and Coulomb), especially at higher

Table 2. Effects of torch angle, weld current, and magnetic field on WAAM bead geometry and surface quality.

Parameter	Effect on bead geometry & surface quality	Remarks
Torch angle (45°–90°)	- Reduces molten metal sagging- Produces stronger bead geometry	Perpendicular orientation preferred; smaller angle improves support during solidification
Weld current	- Increases bead width- decreases penetration depth at high speeds- can cause spatter and reduced reinforcement	High current = more force on molten metal; affects droplet motion and bonding strength
Melting rate factors	- Influenced by wire extension, current flow, and primary current- hot-wire current improves surface quality	Hot-wire technique helps achieve smoother surface morphology
Arc discharge current	- Increases layer diameter- does not affect layer shape	Current mainly alters diameter; critical for volume control
Arc-on time (Ton)	- Affects both layer height and diameter	Must be optimized for uniform layer thickness
Arc voltage	- Minimal effect on layer structure	Less significant than current or Ton
Magnetic field (MF)	- Uniform bead geometry with moderate excitation- >6 A causes arc irregularity and porosity- decreases penetration, increases width	Excessive MF introduces defects due to centrifugal force on the droplet

Table 3. Comparison of material/system, WAAM method, dominant process parameters studied, residual stress (RS) measurement method, qualitative RS severity, and mitigation approach and effectiveness.

Material / build	WAAM process	Key parameters studied	RS measurement	Qualitative RS severity (reported)	Mitigation / control & effectiveness
Ti-6Al-4 V	WAAM + rolling	Rolling pressure, pass schedule	XRD / mechanical testing	**High** (as-built)	**In-process rolling** — **high** effectiveness in texture refinement & RS reduction
Steel alloys	GMAW WAAM	Travel speed, interpass time, heat input	Hole-drilling / FEM	**Moderate–high**	Process parameter optimization — **moderate**; suggested combination with post-treatment
Fe–Fe_3Ni FGM	GTAW WAAM	Interpass temp, current	SEM + strain release	**Moderate**	Thermal control (interpass temp) — **moderate** effectiveness
Ni-based super alloy	WAAM + LSP	Deposition vs LSP schedule	XRD / surface residual stress	**Moderate–high**	**Laser shock peening** — **high** surface RS mitigation (better than shot peening in many reports)
Ni alloys	GMAW WAAM	Travel speed, wire feed rate	Microstructure + porosity measures	**Moderate**	Parameter optimization — **moderate**; recommends coupled controls

travel speeds. High current can cause spatter, decreased reinforcement height, and less metal bonding due to altered heat input. Melting rate is influenced by factors like wire extension and hot-wire current, with hot-wire current improving surface morphology and quality. Arc discharge current affects layer diameter, while arc-on time influences both height and diameter [34-37]. The table below shows the analysis which compares material/system, WAAM method, dominant process parameters studied, residual stress (RS) measurement method, qualitative RS severity, and mitigation approach & effectiveness. Studies were selected to cover common materials (stainless steels, Ni-based alloys, Al alloys).

Across both experimental and modelling studies [38, 39], increasing heat input and poor interpass control correlate with higher residual stress levels. After the parts are manufactured using the additive manufacturing techniques, they are machined in order to achieve the required finished geometries, surface roughness, and shape according to the application. For the part to be more precise, a lot of material has to be removed and due to which the residual stresses are generated in addition to the residual stresses generated during the AM process. One such process to achieve the desired shape is through the milling. During the milling operation, the mechanical and thermal impacts are affecting the workpiece simultaneously and hence, there are three factors that adds to the generation of the residual stresses. The effect is shown in Fig. 6 [40-43].

The chip formation effect creates tensile stresses as the tool will stretch the part at the surface. The smearing effect is the damage caused on the workpiece due to the milling causing the compressive stresses. The heat generated during the milling process causes the workpiece to thermally expand and shrink causing the thermal residual stresses to be generated. The temperature zones and stresses generated due to it are shown in Fig. 7.

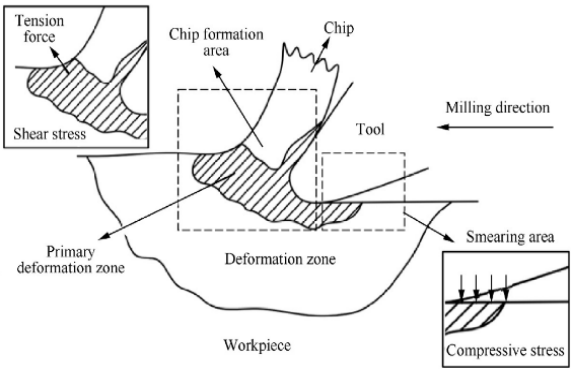


Fig. 6. Mechanical impact of milling process.

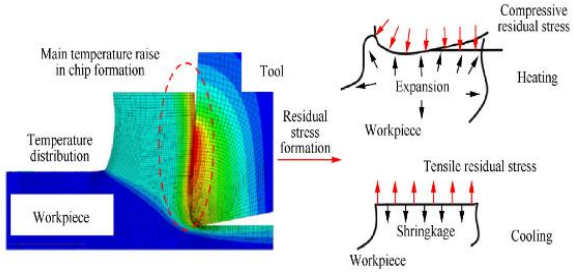


Fig. 7. Stresses due to heat generation.

The heat-input expression used in this study follows the standard formulation for arc-welding and WAAM processes as reported by Kou (2003) and adopted widely in additive manufacturing literature [44, 45]. Hence, we can say that the total residual stress generated in the sample is the combination of thermal and mechanical stresses where chip formation and smearing effect constitute the mechanical stresses [36-39]. Mathematically it can be denoted as,

$$\sigma_{residual} = \sigma_{thermal} + \sigma_{mechanical} = \sigma_{thermal} + (\sigma_{chipping} + \sigma_{smearing})$$

4. Sources and mechanisms of residual stresses in WAAM

Characterization of residual stresses is nothing but the assessment of the defects for the validation of the various mitigation techniques. Residual stress evaluation is a critical aspect in wire arc additive manufacturing (WAAM) due to the severe thermal gradients, cyclic heating–cooling behavior, and constrained solidification associated with layer-by-layer deposition. Residual stress measurement techniques can broadly be classified into destructive, semi-destructive, and non-destructive methods.

Destructive techniques such as sectioning, contour method, and hole-drilling provide direct measurement of internal stresses but are not suitable for in-process monitoring. Among semi-destructive techniques, the hole-drilling strain gauge method is widely used in WAAM studies to estimate near-surface residual stress distributions with acceptable accuracy. Non-destructive techniques are increasingly preferred for WAAM applications because they enable rapid evaluation without damaging the component. X-ray diffraction (XRD) is one of the most commonly used methods for residual stress measurement in metallic WAAM parts. Recent neutron diffraction studies (2024) have achieved higher spatial accuracy in mapping stress fields in WAAM components, providing deeper penetration and improved measurement reliability [46]. These tests can be either be destructive, semi-destructive or non-destructive [47]. The selection of the characterization technique depends upon the elements like material, component size, geometry, and the level of accuracy required for the measurement. Along with this, the accurate modelling of the process is required for understanding the residual stresses and their control. In order to achieve this, understanding the factors like accessibility, resolution, accuracy, and the desired depth of stress penetration is required [48]. Real-time thermal monitoring techniques using embedded sensors have enabled improved tracking of stress accumulation during deposition [49]. These findings support earlier reports while providing enhanced insight into dynamic thermal stress evolution through in-situ measurements. Over the years, the non-destructive techniques have gained significant recognition

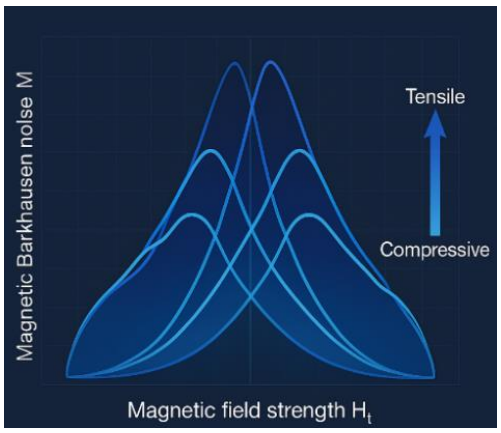


Fig. 8. Stress dependence of the MBN.

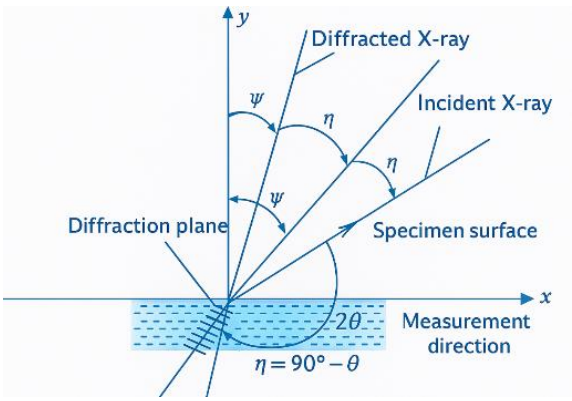


Fig. 9. Diffraction of X-rays.

as it can be used for parts that are at any stage of manufacturing. An advanced non-destructive approach is magnetic barkhausen noise (MBN) analysis, particularly suitable for ferromagnetic WAAM materials such as Fe–Ni alloys. When an alternating magnetic field is applied, the movement of magnetic domain walls generates voltage pulses that are sensitive to microstructural features, hardness variations, and residual stresses. Parameters such as RMS value, peak amplitude, and peak position of the BN signal correlate strongly with stress states, making MBN a rapid and cost-effective technique for surface integrity assessment. The magnetic Barkhausen noise method in which the material is magnetized and the magnetic regions called as domains change the direction in sudden steps due to Bloch walls jumps from one place to another. Bloch walls are the boundaries between the magnetic regions. Due to these jumps of the boundaries, small voltage is created and it is detected by a coil placed near the surface of the material. These signals are termed as Barkhausen noise. The strongest noise happens around a specific point in the magnetization process—near the coercive force (HC)—because that's when most of the jumping happens. We can amplify, filter, and smooth these signals and plot them as a curve, showing how the noise changes with the strength of the magnetic field. These signals are affected by the mechanical stresses generated in the sample. The Fig. 8 shows the signals detected from the Bloch wall jumps [50-55].

Ultrasonic testing is also preferred for many samples as it can test materials up to a wide range of thicknesses. Electromagnetic acoustic transducers (EMATs) use magnetic fields and coils to create ultrasound without needing a couplant. This is useful in dirty or rough environments. But EMATs only work on conductive materials (like metals), and they have a more limited frequency range. Ultrasonic waves are affected by stress in a material. This is called the acousto-elastic effect—as stress changes, the speed of sound waves in the material also changes.

By measuring the time of flight (TOF) of these waves, changes in stress can be detected. A 1 % change in TOF usually means a stress change of about 100 MPa, but this varies by material and wave type. Longitudinal waves (which move in the direction of travel) are generally more sensitive to stress changes than transverse waves [56-60]. Additionally, X-ray diffraction method is also used under the non-destructive methods. This technique is used to measure the stresses at the surface level based on changes in crystal lattice spacing due to stress. By analyzing shifts in diffraction peak angles at various incident angles, strain and corresponding stress can be calculated up to a depth of ~30–40 μm. This technique is limited by the depth penetration and also to the surface conditions like quality, flatness and uneven surfaces. The Fig. 9 shows the diffraction due to the sample surface in various directions and various angles helps calculate the strain which is used in obtaining the stress [61-69].

The destructive and semi-destructive characterization techniques help in assessment of the residual stresses by being easy to carry the tests, and affordable testing apparatuses. These methods are widely standardized and can help in accurately mapping the RS in thicker materials. The damage caused to the samples in these cases are minimal, but it captures average stresses. Compared to conventional methods, MBN offers significant advantages including real-time capability, portability, minimal sample preparation, and easy integration with manufacturing environments. Therefore, it is increasingly being explored as a promising tool for in-situ monitoring and quality control in WAAM processes. These techniques are evolved by combining them with the computational techniques using computers. These techniques are summarized in Table 4 given below [70-78]. Magnetic Barkhausen noise (MBN) analysis has strong applicability to WAAM-fabricated ferromagnetic alloys such as Fe–Ni systems. MBN offers rapid, non-destructive, and real-time stress evaluation with minimal sample preparation. The technique is particularly advantageous for WAAM components because residual stresses are predominantly concentrated near the surface and interlayer regions, which fall within the effective sensing depth of Barkhausen noise measurements. MBN is portable, cost-effective, and easily integrable into manufacturing environments, making it suitable for in-situ monitoring and industrial quality control applications. Its sensitivity to microstructure, hardness, and domain wall pinning behavior allows simultaneous assessment of surface integrity parameters in WAAM builds.

Table 4. The destructive and semi-destructive residual stress measurement methods.

Method	Type	Principle	Typical applications
Hole-drilling	Semi-destructive mechanical	Drill a small hole; measure released strain using strain gauges	Thin plates, welded joints, aerospace components, automotive structural parts, shot-peened surfaces
Contour method	Destructive mechanical	Cut sample; measure deformation of cut surface; calculate stress using elasticity principles	Large welded structures, additively manufactured components, thick-section metals, turbine parts
Ring-/block-cutting	Destructive mechanical	Cut a ring or block section; measure deformation to calculate average residual stress	Pipes, pressure vessels, rolled plates, railway axles, cylindrical components
Indentation	Micro-destructive mechanical	Indent surface; relate indentation shape/strain to residual stress	Heat-treated steels, surface-hardened layers, coatings, weld HAZ regions, AM surface layers
Layer-by-layer milling	Destructive mechanical	Gradually remove layers; measure strain after each removal to assess residual stress	Thick plates, large AM builds, forged blocks, quenched components, welded assemblies

Table 5. Residual stress mitigation techniques in WAAM.

Method	Working principle	Key advantages	Limitations
Stress relief heat treatment (SRHT)	Controlled heating allows plastic relaxation and stress redistribution	Simple, widely used, uniform stress reduction	Grain coarsening, mechanical property degradation if overheated
Hot isostatic pressing (HIP)	High temperature and pressure eliminate internal defects and relax stresses	Removes porosity, improves fatigue life and density	Very expensive, limited to critical components
Shot peening	Surface plastic deformation induces compressive residual stresses	Improves fatigue resistance, low cost	Limited depth effectiveness
Inter-layer rolling	Plastic deformation between deposited layers reduces tensile stresses	Refines grains, improves density, reduces distortion	Requires additional equipment and process control
Hammer peening	Repeated impacts introduce compressive stresses	Portable, effective for large structures	Surface damage risk if uncontrolled
Preheating of substrate	Reduces thermal gradients during deposition	Prevents cracking, improves bonding	Increases heat accumulation risk
Interpass temperature control	Maintains uniform thermal conditions between layers	Reduces thermal stress buildup	Slows production rate
Optimized deposition path planning	Balanced heat distribution through strategic toolpath	Reduces distortion, improves dimensional accuracy	Complex programming required
Ultrasonic vibration assistance	Ultrasonic energy refines microstructure and reduces stress accumulation	Improves grain refinement and mechanical properties	Equipment complexity
Laser-assisted WAAM	Secondary heat source controls cooling rate and thermal gradients	Reduces residual stress and improves fusion quality	High cost and complexity

5. Post processing and mitigation strategies for residual stresses and defects in WAAM

After the sample is prepared, it is processed in order to in order to improve its micro-structural and in turn its mechanical properties. New findings highlight the continued effectiveness of rolling-assisted WAAM in reducing residual stress by inducing controlled plastic deformation [79]. Laser shock peening (LSP) has emerged as a strong alternative to shot peening, showing significant improvements in stress relief for nickel-based WAAM components [80]. Commonly, heat treatment is performed on the sample to improve the mechanical properties of the sample produced. Due to the heat treatment, the stresses in the sample are relieved. The sample is heated in the furnace below its critical temperature and kept in the furnace for a certain amount of time. This releases the internal stresses within the sample. Similarly, aging of the sample also helps in reducing the residual stresses as while the sample is heated, it is also cooled in a controlled manner. This treatment allows the precipitation of secondary phases. This prevents the dislocation of the by avoiding the sliding of atoms in the grain structure, which helps in increasing the material's strength, creep resistance, and fatigue life [66-69]. Rolling is another process in which high pressure rollers are used which prevents the development of extensive columnar grains by the process of recrystallization. The limitation of this process is inconsistent recrystallization level, costly equipment, increased deposition time and limited complexity in the geometry of the part [81-85].

Another method is by using shot peening, the part's surface can be strengthened, fatigue resistance can be improved and porosity can be reduced. In this process, small sized balls of either metal, ceramic or glass is bombarded on the surface of the material. This reduces the internal stresses by 4 % to 6 % near

the surface and improved the mechanical properties of the metal. Shot peening makes the surface rough but polishing improves the roughness and increases the fatigue resistance. The low intensity shot peening will create better surface finish as compared to the high intensity [86-90]. Additionally, post-deposition heat treatment has been shown to refine microstructure, reduce residual stresses, and stabilize precipitate formation in super alloys [91]. These new results strengthen the case for integrated post-processing strategies.

Tensile properties obtained from WAAM components are strongly influenced not only by heat input and deposition parameters but also by surrounding environmental conditions such as ambient temperature, humidity, airflow, and shielding-gas stability. Slight variations in shop-floor temperature can alter arc energy density and cooling rates, while high humidity or unstable shielding gas flow increases oxidation, porosity, and arc wandering, leading to inconsistent bead geometry and degraded tensile strength. Studies on TIG welding of aerospace aluminum alloys demonstrate that fluctuations in arc length, gas purity, and ambient conditions produce measurable changes in joint strength, ductility, and weld penetration. These parallels indicate that WAAM tensile performance is sensitive to comparable physical phenomena-arc stability, melt-pool behavior, and oxidation dynamics. Therefore, controlling the deposition environment (temperature ~20–25 °C, low humidity < 40 %, laminar airflow, and consistent shielding gas flow) is essential to ensure reproducible tensile properties and reduce scatter in mechanical performance evaluation. Wire arc additive manufacturing (WAAM) is inherently characterized by high heat input, steep thermal gradients, and repeated thermal cycling, which lead to the development of significant residual stresses, distortion, and microstructural heterogeneity. To improve structural integrity and dimensional accuracy, several post-processing and in-process mitigation strategies have been developed.

6. Current challenges and future scope in WAAM

WAAM is one of the very efficient and affordable technique and has gained much attention in recent years. In order to achieve its full potential, on line monitoring or in situ monitoring systems are very much essential. Using this, the stresses can be detected while the part is being produced and also the process parameters can be modified simultaneously. This will help in reduction in the generation of the residual stresses. A near defect free production of the parts can be obtained if automation in the system is done by using the robots for integrated real time monitoring and quality control systems [92-96]. Machine-learning-assisted WAAM process optimization has demonstrated promise for predicting bead height, width, and thermal behavior based on prior deposition history [97]. Considering the ongoing efforts in reduction of the cost of manufacturing and producing bulk quantities of the samples, the uneven dissemination of microstructure, anisotropy in mechanical properties, high residual stresses and distortion of parts are few of the major challenges faced by the additive manufacturing industry [98-100]. In order to increase the efficiency of the production process, the machining platforms can be integrated and by using a lighter but stronger material will lead to a drastic boost in the usage of this technique. A skilled workforce will add a lot of value in this technique by bridging the gap between designs and manufacturing [101]. Digital twin approaches are enabling new possibilities for in-process thermal control and dynamic path adjustment. These innovations collectively represent the forward trajectory for intelligent, automated WAAM systems. Laser technology plays a significant role in modern manufacturing, particularly in material processing, hybrid additive manufacturing, and residual stress mitigation. In laser-based processes, a highly concentrated coherent energy beam interacts with the material, resulting in localized melting, vaporization, or thermal modification. The primary advantage of laser processing lies in its high energy density, precise heat input control, and minimal heat-affected zone (HAZ), which collectively enable improved dimensional accuracy and surface integrity. In the context of WAAM and hybrid additive manufacturing systems, laser technology is commonly employed as an auxiliary heat source to control thermal gradients during deposition. By providing localized reheating, laser assistance helps reduce cooling rates between successive layers, thereby minimizing residual stress formation, distortion, and anisotropic microstructural evolution. Laser remelting has also been reported to refine grain structures, improve interlayer bonding, and enhance mechanical properties through homogenization of elemental distribution [102].

Laser cutting and laser–material interaction studies further demonstrate the influence of processing parameters on surface quality and dimensional characteristics. Parameters such as laser power, scanning speed, beam focus position, assist gas pressure, and cutting geometry significantly affect kerf width, surface roughness, and thermal damage. Studies on laser machining of X60 steel indicate that higher cutting speeds reduce heat input and improve surface finish, while increased laser power tends to enlarge kerf width due to excessive melting. Optimization of these parameters is therefore critical to achieving superior surface integrity.

In WAAM applications, integrating laser technology offers several advantages, including improved thermal management, reduction of tensile residual stresses, enhanced metallurgical bonding in multi-material deposition, and real-time defect mitigation. However, challenges such as high equipment cost, complex process synchronization, and energy efficiency considerations remain key limitations for widespread industrial implementation [103-108].

7. Conclusions

This study comprehensively reviewed the formation mechanisms, characterization techniques, and mitigation strategies of residual stresses in wire arc additive manufacturing (WAAM) processes, with particular emphasis on multi-material and twin-wire deposition systems. Residual stresses in WAAM primarily

originate from steep thermal gradients, cyclic heating–cooling during layer-by-layer deposition, solidification shrinkage, phase transformations, and mechanical constraints imposed by substrate and previously deposited layers. These stresses typically manifest as tensile stresses along the build direction and compressive stresses near the substrate, leading to distortion, cracking susceptibility, anisotropic mechanical properties, and compromised structural integrity. The analysis highlighted that residual stress magnitude and distribution are strongly influenced by process parameters such as heat input, interpass temperature, travel speed, wire feed rate, and deposition strategy. Multi-material WAAM further intensifies residual stress development due to thermal expansion mismatch and metallurgical incompatibility at the interfaces. Twin-wire WAAM offers potential advantages in controlling dilution, compositional gradients, and heat input distribution, thereby providing opportunities for improved stress management.

To obtain an optimal geometrical accuracy and material properties in the samples prepared using WAAM, heat input, travel speed, wire feed rate, interpass temperature, and deposition strategy have the most significant influence on residual stress evolution. Studies consistently indicate that high heat input and lower travel speeds increase tensile longitudinal stresses, whereas controlled interpass temperature (≈ 150–250 °C), shorter weld lengths, and oscillation-assisted deposition help reduce gradients and stabilize bead morphology. It has been observed that the bead height is more important than the width and it's binding over the previous layer having peak current. Characterization techniques such as neutron diffraction and X-ray diffraction remain the most accurate for bulk and surface stress assessment, respectively, while ultrasonic TOF and Barkhausen noise offer efficient nondestructive monitoring for in-process evaluation. Post-processing methods such as heat treatment, rolling, and laser shock peening show measurable reductions in residual stress. The parts manufactured using the WAAM showed equal or better properties than the conventional wrought or cast products. Overall, the findings indicate that optimizing process parameters at the deposition stage—especially heat input, interpass temperature, and deposition strategy—combined with targeted post-processing, is the most effective route to improve dimensional stability and mechanical performance in WAAM components. More penetrative research works should be carried out to analyze fatigue and corrosion properties of WAAM super alloys for future industrial applications. Additionally, process parameters optimization and better control over quality are the other aspects to be focused by further studies.

The review also demonstrated that residual stress mitigation in WAAM can be effectively achieved through a combination of process optimization, thermal management, mechanical post-processing, and hybrid manufacturing approaches. Techniques such as interlayer rolling, laser shock peening, machine hammer peening, heat treatment, and laser-assisted WAAM significantly reduce tensile residual stresses, refine microstructures, improve grain boundary characteristics, and enhance mechanical performance. Among these, hybrid laser-arc approaches show considerable potential by enabling controlled heat input, reduced thermal gradients, and improved interlayer bonding. Overall, the findings establish that an integrated framework combining optimized process parameters, advanced post-processing methods, and real-time non-destructive monitoring techniques is essential for controlling residual stresses and ensuring structural reliability in WAAM components. The study provides valuable guidelines for developing scalable, high-quality WAAM processes, particularly for multi-material and functionally graded metallic structures intended for critical engineering applications.

One of the most critical future directions involves the development of real-time residual stress monitoring systems integrated directly within WAAM setups. Current measurement techniques are largely post-process and laboratory-based, which limits their applicability in industrial environments. Future research should focus on embedding non-destructive evaluation tools such as magnetic Barkhausen noise (MBN), acoustic emission sensors, and thermographic monitoring into WAAM platforms to enable in-situ stress detection and adaptive process control. From an industrial perspective, there is a pressing need to establish standardized protocols and guidelines for residual stress measurement, acceptance limits, and qualification of WAAM components, particularly for critical sectors such as aerospace, nuclear, cryogenic, and defence applications. Development of internationally accepted standards will accelerate technology transfer from laboratory research to industrial manufacturing.

Finally, future work should also emphasize correlating residual stress with long-term performance metrics, including fatigue life, creep behaviour, corrosion resistance, and fracture toughness of WAAM components. Such studies will provide a comprehensive understanding of structural reliability under service conditions.

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Conflict of interest

The authors declare no competing interests.

Nomenclature

Abbreviations

WAAM : Wire arc additive manufacturing
AM : Additive manufacturing
GTAW : Gas tungsten arc welding
GMAW : Gas metal arc welding
PAW : Plasma arc welding
SEM : Scanning electron microscopy
MBN : Magnetic Barkhausen noise

EMAT : Electromagnetic acoustic transducer
TOF : Time of flight
XRD : X-ray diffraction
FGM : Functionally graded material
MF : Magnetic field
RS : Residual stress
FE : Finite element
LSP : Laser shock peening

Symbol

I : Welding/deposition current (A)
V : Arc voltage (V)
v : Travel speed (mm/min or mm/s)
WFR : Wire feed rate/speed (mm/s or mm/min)
h : Layer height / bead height
d : Bead width
a : Arc length (distance between electrode and workpiece)
θ : Torch angle
T : Temperature (°C)
Tint : Interpass temperature
Ton : Arc-on time
Q : Heat input
ρ : Material density
k : Thermal conductivity
α : Coefficient of thermal expansion
σ : Stress (general symbol)

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